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Review

Effects of isometric vs. dynamic resistance training on isometric and isokinetic muscular strength: A systematic review with meta-analysis☆

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ABSTRACT

Backgrounds: Both isometric resistance training (ISO-RT) and dynamic resistance training (DYN-RT) are commonly used interventions to improve muscular strength. Unlike DYN-RT, which involves producing force through a joint range of motion, ISO-RT generates force without joint movement. However, their relative effectiveness for improving strength remains unclear, making it essential to clarify these differences to better guide exercise prescription across diverse populations and training goals.
Objectives: This systematic review and a meta-analysis investigated the effects of ISO-RT versus a non-exercise control (CTRL) or DYN-RT on isometric and isokinetic muscular strength in healthy adults.
Methods: Four databases (PubMed/MEDLINE, SPORTDiscus, Embase, and Scopus) were searched up to May 2025. Randomised and non-randomised controlled trials, including adults (18–45 years), were eligible for inclusion. Random-effects meta-analyses were conducted, with outcomes reported as standardised mean differences (SMD) and 95 % confidence intervals (CIs).
Results: A total of 32 studies involving 621 participants were included. ISO-RT significantly improved combined isometric and isokinetic strength compared to CTRL (SMD = 0.65; 95 % CI: 0.52–0.77; $p < 0.0001$) and DYN-RT (SMD = 0.35; 95 % CI: 0.21–0.48; $p < 0.0001$). Subgroup analyses revealed greater isometric strength gains with ISO-RT compared to DYN-RT (SMD = 0.43; $p < 0.0001$), whilst no statistically significant difference was observed for isokinetic strength between these RT modalities (SMD = −0.20; $p = 0.24$).
Conclusions: Overall, our findings indicate that ISO-RT resulted in greater isometric strength gains compared to DYN-RT and CTRL, whilst isokinetic strength gains were similar between these RT modalities. These findings indicate that ISO-RT may be an effective modality for enhancing muscular strength and could be considered a viable alternative or complement to DYN-RT.
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Practical implications

For practitioners aiming to improve isometric and isokinetic strength, the evidence supports using ISO-RT protocols involving high-intensity contractions (≥70–100 % MVC) performed for 3–10 s per repetition, across 3–5 sets, 2–3 times per week. Training at one to three joint angles per exercise is common practice, although multi-angle approaches may offer enhanced transfer to dynamic movements. Whilst most of the evidence base is derived from untrained populations, these parameters provide a practical foundation for programme design. ISO-RT can be especially useful in clinical, rehabilitative, or joint-specific training scenarios where dynamic movement may be contraindicated.

1. Introduction

Muscular strength is a key contributor to athletic performance, physical function, and overall health, with established associations with reduced all-cause mortality in healthy populations.[1,2] Strength is commonly assessed using isometric, dynamic, or isokinetic tests, each capturing different aspects of neuromuscular function.[3–5] Isometric tests evaluate maximal force at a fixed joint angle, while dynamic tests assess strength through a range of motion, and isokinetic tests measure torque at controlled angular velocities.

Resistance training (RT) is widely prescribed to enhance muscular strength in both athletic and general populations.[2,6] Two commonly

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applied RT modalities are dynamic RT (DYN-RT) and isometric RT (ISO-RT), each producing distinct neuromuscular adaptations.[7] DYN-RT involves concentric and eccentric muscle actions across a joint range of motion and is typically prescribed to enhance functional strength in movement-based tasks. In contrast, ISO-RT consists of static muscle action at fixed joint angles, with no visible change in the muscle-tendon unit.[8]

ISO-RT typically produces joint-angle strength gains and may be especially useful for targeting force production at biomechanically challenging positions or when joint movement is restricted, such as in rehabilitation.[9] Although DYN-RT is more widely used, ISO-RT has unique advantages. The absence of joint movement reduces mechanical stress, making ISO-RT suitable for individuals with joint pathologies or mobility limitations.[10–13] ISO-RT also avoids eccentric loading, which lowers muscle damage and may allow higher training frequency and volume, potentially enhancing strength and hypertrophy.[11] Isometric muscle actions can also produce 15–20 % greater peak force than concentric actions, increasing neuromuscular activation and mechanical tension, especially when performed at longer muscle lengths, thereby further supporting strength and hypertrophic adaptations.[12]

Despite these benefits, ISO-RT is sometimes considered limited due to its joint-angle specific effects and exclusion of eccentric and concentric muscle actions, which contribute to hypertrophy.[12,14] In DYN-RT, muscles are loaded through a full range of motion, enhancing mechanical tension across varying lengths and joint angles, factors associated with greater muscle growth.[8] However, recent evidence suggests that training at longer muscle lengths,[13] and using multiple joint angles in ISO-RT may broaden its benefits and improve transfer to dynamic tasks such as the 1RM bench press.[15] This highlights the importance of joint angle selection and training configuration in maximising ISO-RT outcomes.[8,10]

Similar to DYN-RT, ISO-RT allows systematic manipulation of key training variables such as contraction intensity (typically expressed as a percentage of maximum voluntary contraction - MVC), time under tension, and joint angle.[8,10,16] Several studies have reported significant improvements in MVC, rate of force development, and even performance in dynamic activities, like sprinting, following ISO-RT.[17–20] However, the relative efficacy of ISO-RT compared to DYN-RT remains unclear. Some studies report greater strength gains following DYN-RT,[21,22] whilst others have found similar improvements between the two RT modalities.[15,23,24] These differences likely reflect the principle of task specificity, whereby the greatest improvements occur in the trained muscle action mode.[25] Isokinetic strength testing, a muscle action mode distinct from both RT types, may serve as a neutral comparator, but its use in comparative research has been limited.[26]

To date, no meta-analysis has systematically compared the effects of ISO-RT versus DYN-RT on strength outcomes. Prior reviews, such as Oranchuk et al.[8] focused on the role of muscle length during ISO-RT, but did not assess its efficacy relative to DYN-RT. A narrative review by Lum and Barbosa[10] supported the benefits of ISO-RT but lacked systematic methodology, potentially introducing bias. The aim of this study was to systematically review and meta-analyse the effects of ISO-RT on muscular strength, compared to non-exercise controls (CTRL) and DYN-RT, in healthy individuals. Subgroup and meta-regression analyses also explored the influence of training variables, with the goal of informing evidence-based application of ISO-RT in strength programming.

2. Methods

2.1. Reporting and registration

The systematic review conformed to the "Preferred Reporting Items for Systematic Reviews and Meta-Analyses" (PRISMA) guidelines.[27] Therefore, no Institutional Review Board approval was necessary. The complete study protocol was registered on PROSPERO (ID: CRD42024558772). Whilst the original protocol included plans to evaluate muscular hypertrophy as a secondary outcome, this component was omitted from the final analysis due to insufficient eligible studies reporting hypertrophy using standardised, quantifiable methods. As such, the scope of the review was refined to focus exclusively on muscular strength outcomes.

2.2. Literature search methodology

A comprehensive search was created and conducted in the following databases: PubMed/MEDLINE, SPORTDiscus, Embase, and Scopus, covering all records from inception to June 2024. The search was updated in May 2025. The searches were language-restricted to English. Forward and backward citation checking of relevant articles was performed to ensure comprehensive coverage. Key terms were searched within the article title, abstract, and keywords using conjunctions "OR" and "AND" with truncation "*". The search terms included: "isometric training" OR "isometric contraction" OR "isometric training" OR "isometric" "isometric contraction" OR "static contraction". These terms were then combined with "strength" OR "isometric strength" OR "isokinetic strength".

2.3. Inclusion and exclusion criteria

Studies were included if they were randomised or non-randomised comparative trials, published in peer-reviewed English-language journals, involved healthy adults aged 18–45, and reported changes in isometric or isokinetic strength. Eligible studies included at least one group that completed ISO-RT alone and a time-matched comparator group—either a CTRL or a DYN-RT group. Studies were excluded if they involved pathological populations, used isolated finger, toe, wrist, or neck exercises, included bodyweight-only training, combined ISO-RT with other exercise modalities or supplementation, or lacked a comparator group. Two reviewers independently screened abstracts and full texts against these criteria. Disagreements were resolved via a consensus meeting with a third reviewer.

Both within-subject (contralateral limb control) and between-subject (independent control group) designs were eligible to improve statistical power, estimate precision, and generalizability. When both designs were used in a single study, data from the independent control group were prioritised to minimise bias from cross-education effects.[28] Included DYN-RT comparators were limited to traditional isotonic RT (concentric-only or combined concentric-eccentric). Studies involving eccentric-only protocols were excluded due to insufficient data, ensuring consistency in training modalities and avoiding confounding from differing mechanical demands.

2.4. Data extraction

Two independent reviewers (M.G., J.L.) conducted a systematic data extraction process using a predefined and standardised spreadsheet. Any reviewer discrepancies were discussed and resolved by consensus; unresolved differences were referred to a third reviewer (D.H.) for adjudication. The extracted data included participant demographics (e.g., age, sex, training status), details of intervention programming (e.g., sets, repetitions, inter-set rest duration, and relative load expressed as %1RM for ISO-RT group only), exercise characteristics (e.g., type and number of exercises), intervention specifics (e.g., duration and frequency), ISO-RT loading details (e.g., MVC% and time under tension), and supervision status. Additionally, data on absolute performance values (mean ± standard deviation) for primary and secondary outcome measures were gathered for pre- and post-intervention assessments. These outcomes included measurements for

muscle strength (e.g., isometric and isokinetic tests for strength). Where graphical data were present,[21,29,30] values were digitised using WebPlotDigitizer version 4.1.0 (WebPlotDigitizer, San Francisco, CA).

2.5. Methodological quality and risk of bias

Two reviewers (M.G., M.K.) independently conducted quality assessments of the included studies, resolving disagreements through a consensus meeting. The methodological quality was evaluated using the Tool for the Assessment of Study Quality and Reporting in Exercise (TESTEX),[31] a validated tool that includes a 15-point scale designed to assess both study quality (5 points) and reporting (10 points). Each criterion was scored as '1' if met and '0' if not. The study quality assessment focused on randomisation (e.g., coin-tossing), allocation concealment (concealment before randomisation), baseline comparability, and assessor blinding. Although the TESTEX does not provide specific instructions for classifying studies based on their scores, a consensus was reached to categorise studies as follows: a total score below 4 was classified as "poor," scores between 4 and 7 were considered "moderate," scores from 8 to 11 were deemed "good," and scores above 11 were labelled as "excellent." The risk of bias was assessed through visual inspection of funnel plots (Supplementary Figs. 1 & 2) and further evaluated using Egger's regression asymmetry test to identify potential publication bias. In cases where significant asymmetry was detected with Egger's test, the Trim and Fill method was applied to adjust for potential bias.[32]

2.6. Statistical analysis

Each study was independently analysed for the primary outcome of muscle strength, with pre-and post-intervention means and standard deviations extracted and compiled. Standardised mean differences (SMDs) with 95 % confidence intervals (CIs) were calculated to account for variability in measurement tools across studies.[33] For outcomes with more than five studies, prediction intervals (PIs) were also computed using a t-distribution to reflect between-study heterogeneity and provide an estimate of likely effects in future studies.[34]

Primary analyses compared ISO-RT to CTRL and DYN-RT. Subgroup meta-analyses were performed based on body region (upper vs. lower body), contraction type during testing (isometric vs. isokinetic), and study duration (≤6 vs. >6 weeks), provided at least two studies were available per subgroup. Because the pre-post correlation coefficient (r) was not reported in the included studies, we assumed a value of 0.5. This value is commonly used in meta-analyses when pre-post correlations are not reported, as it lies within the recommended range of 0.5 to 0.7, suggested by Follmann et al.[35] and Rosenthal et al.[36] We selected $r = 0.5$ as a conservative estimate to reduce the risk of overestimating precision and inflating false-positive findings (Type I error).[37] In studies assessing strength across multiple joint angles or muscle action types, separate effect sizes were computed for each condition. A two-level random-effects model was employed to account for multiple outcomes nested within studies, thereby enhancing robustness and minimising bias. SMDs were interpreted using the following thresholds: trivial (≤0.20), small (0.21–0.50), moderate (0.51–0.80), and large (>0.80). Positive values favoured ISO-RT, whilst negative values favoured CTRL or DYN-RT. Statistical heterogeneity was assessed using the Q test and I^2 index, with I^2 values between 30 % and 60 % indicating moderate heterogeneity. A p-value of <0.05 was considered statistically significant.

2.7. Meta-regression

Mixed-effects meta-regression analyses were conducted to examine the potential influence of specific ISO-RT moderators on muscle strength across studies. The moderators analysed included training intensity (expressed as MVC%), training volume (number of sets and repetitions per set), time under tension, and rest intervals between sets. For categorical moderators, such as testing load and body region assessed, binary coding (e.g., "1" and "2") was applied. In cases where studies provided a range of repetitions for the ISO-RT prescription, the median value within the range (e.g., 8–12 repetitions = 10 repetitions) was used in the analyses. Each meta-regression used a mixed-effects model to assess the relationship between these moderators and effect size. All statistical analyses were conducted using the Meta package in the R 4.3.2 programming environment.

2.8. Sensitivity analysis

A leave-one-out sensitivity analysis was conducted to evaluate the robustness of the pooled effect estimates. This approach involved systematically omitting one study at a time and recalculating the overall effect size to assess whether any single study unduly influenced the results. All analyses were performed using a random-effects model. Stability of the SMDs and confidence intervals across iterations were used to indicate the reliability of the meta-analytic findings. This procedure was applied to both primary analyses comparing strength gains following ISO-RT versus CTRL and ISO-RT versus DYN-RT.

2.9. Publication bias

Publication bias was conducted using the "Trim and Fill" method to evaluate the robustness of our meta-analytic findings and assess potential publication bias. This procedure identifies and adjusts for funnel plot asymmetry by estimating the number of potentially missing studies and recalculating pooled effect sizes after imputing these hypothetical studies. This helps determine whether the observed findings remain stable after accounting for publication bias.

3. Results

3.1. Included studies

After removing duplicates, 12,158 records were screened from electronic databases. A total of 75 articles were assessed for eligibility, with 32 studies meeting the inclusion criteria (Fig. 1).[18–24,29,30,38–60] A total of 621 participants were included across all studies. Of these, 467 (75.2 %) were males, and 135 (21.8 %) were females, with one study not reporting sex distribution.[27] The mean age of participants ranged from 19 to 38 years. Twenty-four studies included untrained participants, whilst eight studies involved trained participants. Strength gain was assessed via isometric tests in 27 studies; one study measured strength using isokinetic tests, and four studies utilised both isometric and isokinetic tests to measure strength gain. A detailed summary of study characteristics is provided in Table 1.

3.2. Strength

Twenty-eight studies, encompassing 69 effects, compared strength gains following ISO-RT with CTRL. ISO-RT resulted in significant improvements in muscular strength compared to CTRL, as assessed by both isokinetic and isometric strength tests (SMD = 0.65, $p < 0.0001$, 95 % CI = 0.52 to 0.77, 95 % PI = 0.52 to 0.78; $I^2 = 0$ %) (Fig. 2). Although Egger's test (intercept = −0.19, p = 0.01) indicated potential publication bias, the Trim and Fill method showed no change in the effect size, confirming the robustness of the results. Seven studies compared ISO-RT with DYN-RT, yielding 33 effect sizes. ISO-RT also produced significant improvements in muscular strength compared to DYN-RT (SMD = 0.35, $p < 0.0001$, 95 % CI = 0.21 to 0.48, 95 % PI = 0.21 to 0.49; $I^2 = 0$ %) (Fig. 3). Furthermore, Egger's test (intercept = 0.12, p = 0.25) and funnel plots indicated no evidence of publication bias (Supplementary Figs. 1 & 2).

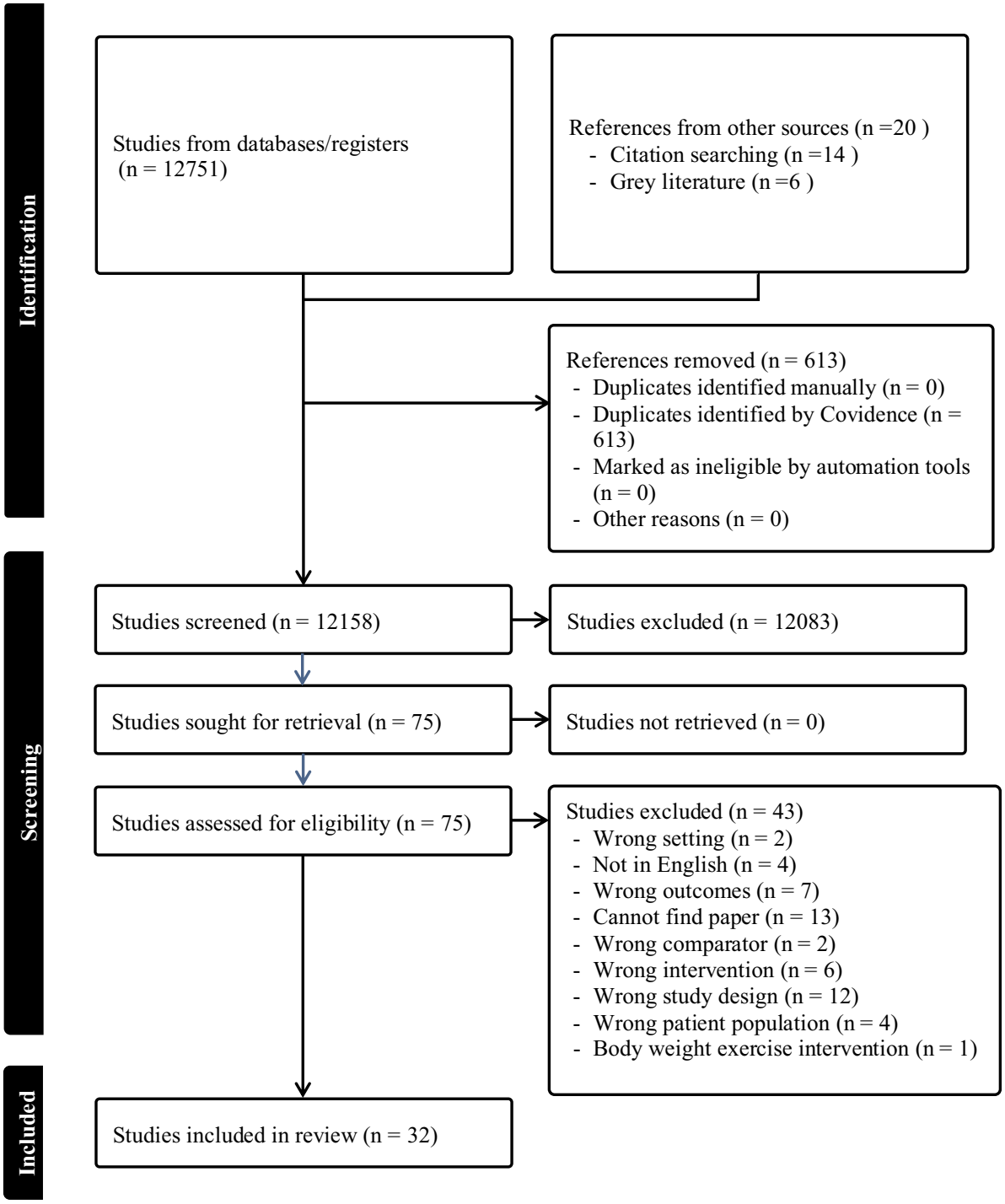

Fig. 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram of the selection process for studies included.

3.2.1. Strength: sub-analyses

3.2.1.1. Upper and lower body. For ISO-RT vs. CTRL, significant improvements were found for both upper body (SMD = 0.71, $p < 0.0001$, 95 % CI = 0.51 to 0.91) and lower body strength (SMD = 0.62, $p < 0.0001$, 95 % CI = 0.45 to 0.78; $I^2 = 0$ %). For ISO-RT vs. DYN-RT, lower body strength significantly improved in ISO-RT (SMD = 0.36, $p < 0.0001$, 95 % CI = 0.23 to 0.50; $I^2 = 0$ %), with no publication bias (Egger's test: intercept = 0.11, p = 0.20). A meta-analysis for upper body strength was not conducted due to insufficient studies.

3.2.1.2. Isometric and isokinetic. ISO-RT showed significant improvements in both isometric (SMD = 0.66, $p < 0.0001$, 95 % CI = 0.52 to 0.80) and isokinetic strength (SMD = 0.23, p = 0.0002, 95 % CI = 0.23 to 0.78; $I^2 = 0$ %) compared to CTRL. Egger's test indicated no publication bias for either isometric (intercept = 0.20, p = 0.22) or isokinetic outcomes (intercept = 0.04, p = 0.89).

Compared to DYN-RT, ISO-RT produced a significant improvement in isometric strength (SMD = 0.43, $p < 0.0001$, 95 % CI = 0.27 to 0.59; $I^2 = 0$ %). However, publication bias was detected (Egger's intercept = −0.05, p = 0.004). After applying the Trim and Fill method, the effect size slightly decreased but remained significant

Table 1
Characteristics of the studies investigating ISO-RT on muscle strength.

Study	Group	Subjects	Prescription	Frequency	Duration	Assessment/s
Balshaw et al.[39]	EISO-RT (EC)	13 UT (25.0 ± 2.0 y), M	EISO KE (UNI): 4 sets × 10 rep @ 80 % MVC (1 s per rep), 120 s rest b/w sets	3	12	ISO-KE; MV
	ISO-RT (SC)	16 UT (25.0 ± 2.0 y), M	ISO KE (UNI): 4 sets × 10 rep @ 75 % MVC (4 s per rep), 120 s rest b/w sets	3	12	
	CTRL	14 UT (25.0 ± 3.0 y), M	NA	–	–	
Balso et al.[40]	ISO-RT (PF)	10 UT (23.8 ± 4.8 y), M	ISO PF: 6 sets × 10 rep @ 100 % MVC (3 s per rep), 120 s rest b/w sets	3	4	ISO-PF
	CTRL	10 UT (22.8 ± 3.0 y), M	NA	–	–	
Carr et al.[41]	ISO-RT (EF)	15 UT (24.0 ± 4.0 y), F	ISO EF: 5 sets × 5 rep @ 80 % MVC (5 s per rep), 90 s rest b/w sets	2	4	ISO-EF
	CTRL	10 UT (25.0 ± 3.0 y), M/F	NA	–	–	
Colson et al.[42]	ISO-RT (EF)	8 UT (22.9 ± 3.0 y), M	ISO EF: 5 sets × 6 rep @ 90 % MVC (6 s per rep), 180 s rest b/w sets	3	7	ISK_EF
	CTRL	8 UT (20.6 ± 2.2 y), M	NA	–	–	
Davies et al.[43a]	ISO-RT (EF)	12 UT (28.5 ± 6.8 y), M/F	ISO KE: 4 sets × 6 rep @ 90 % MVC (4 s per rep), 60 s rests b/w sets	3	6	ISO-EF; CSA
Durbaba et al.[44]	ISO-RT (KE)	7 UT (28.5 ± 6.8 y), M	ISO KE: 15 sets × 10 rep @ 60 % MVC (4 s per rep), 90 s rest b/w sets	3	4	ISO-KE
	CTRL	7 UT (28.5 ± 6.8 y), M/F	NA	–	–	
Fisher et al.[46]	ISO-RT (DL)	8 UT (21.0 ± 1.3 y), M	ISO DL: 6 sets × 1 rep @ 100 % MVC (35 s per rep), 60 s rest b/w sets	2	6	ISO-DL
	CTRL	8 UT (21.1 ± 0.7 y), M	NA	–	–	
Fimland et al.[45]	ISO-RT (PF)	15 T (24 ± 2.0 y), M/F	ISO PF: 6 sets × 6 rep @ 55 % (4 s per rep), 120 s rest b/w sets	4	4	ISO-PF
	CTRL	11 T (24.0 ± 2.0 y), 5 M,6F				
Folland et al.[23a]	ISO-RT (KE)	33 UT (21.5 ± 2.1 y), M	ISO KE (UNI): 4 sets × 10 rep @ 75 % MVC (2 s per rep), 120 s rest	3	9	ISO & ISK-KE
	DYN-RT	33 UT (21.5 ± 2.1 y), M	DYN KE (UNI): 4 sets × 10 rep @ 75 % (1 s CON, 1 s ECC) MVC120 sec rest			
Garfinkel et al.[47]	ISO-RT (KE)	8 UT (21.9 ± 2.7 y), F	ISO KE: 3 sets × 10 rep @ 100 % MVC (4 s per rep)	3	8	ISO-KE; CSA
	CTRL	7 UT (21.9 ± 2.7 y), F	NA	–	–	
Herbert et al.[48]	ISO-RT (EF)	18 UT (age NR), M/F	ISO EF: 6 sets × 1 rep @100 % MVC (10 s per rep) 60 s rest b/w sets	3	8	ISO-EF
	CTRL	18 UT (age NR), M/F	NA	–	–	
Jones et al.[24a]	ISO-RT(KE)	6 UT (27.5 ± 5.7 y), M	ISO KE (UNI): 4 sets × 6 rep @ 80 % MVC (4 s per rep), 60 s rest b/w sets	3	12	ISO-KE; CSA
	DYN-RT	6 UT (27.5 ± 6.3 y), M/F	CON KE (UNI): 4 sets × 6 rep @ 80 % MVC,60 s rest b/w sets			
Kubiak et al.[49]	ISO-RT (KE)	10 UT (18–30 y), M/F	10 sets × 1 rep @ 100 % MVC (10 s per rep), 50 s rest b/w sets	3	5	ISO & ISK-KE & KF
	CTRL	9 UT (18–30 y), M/F	NA	–	–	
Lanza et al.[18]	ISO-RT (KE)	13 UT (22.0 ± 3.0 y), M	3 sets × 6 rep @ 100 % 1RM (3 s per rep), 180 s rest b/w sets	3	4	ISO-KE
	CTRL	9 UT (23.0 ± 3.0 y), 9 M	NA	–	–	
Lee et al.[22]	ISO-RT (KE)	11 UT (21.7 ± 2.8 y), M	ISO KE: 4 sets × 10 rep @ 75 % MVC (1 s per rep), 60 s rest b/w sets	3	8	ISO & ISK-KE
	DYN-RT (KE)	10 UT (21.4 ± 2.9 y), M	DYN KE: 4 sets × 10 rep @ 75 % 1RM (tempo: 1-0-1), 60 s rest b/w sets			
	ISK-RT (KE)	10 UT (22.0 ± 3.4 y), M	ISK KE: 4 sets × 10 rep @ 75 % MVC (90°/s)			
Lum et al.[59]	ISO-RT(MTP & PF)	9 T (36.7 ± 6.0 y), M/F	ISO MTP & PF: (2–3) sets × (3–5) rep @ 100 % MVC (3 s per rep), 180 s rest b/w sets	2	6	ISO-MTP
	DYN-RT	8 T (32.0 ± 7.3 y), M/F	CT: BW (SQ, LN & SL-RDL) 3 sets × 30 s, 30 s rest b/w sets			
	PLY	9 T (37.7 ± 6.6 y), M/F	PLY (DJ & SLB & SJ): 2–4 sets × 5 rep, 180 s rest b/w sets			
Lum et al.[15,50]	ISO-RT (SQT)	9 T (23.0 ± 2.7 y), M	Regular soccer conditioning plus ISO_SQT: 2 sets × 5 rep @ 100 MVC (3 s per rep)	2	24	ISO-MTP
	CTRL	6 T (23.0 ± 2.7 y), M	NA	–	–	
Lum et al.[60]	ISO-RT	10 T (22.0 ± 4.0 y), M/F	Same as DYN-RT in addition to ISO (SQ, BP & PBP): 1–4 sets × 4–8 rep @ 100 % MVC (3 s per rep)	2	6	ISO-BP, SQ & PBP
	DYN-RT	10 T (21.0 ± 4.0 y), M/F	DYN RT (SQ, BP, BP, PU, SP & SR): 2–4 sets × 4–8 rep @ 80–85 % of 1RM			
McDonagh et al.[51]	ISO-RT (EF)	4 UT (19–26 y), M	ISO EF: 10 sets × 30 rep @ 100 % MVC (4 s per rep) 20 s rest b/w sets	7	5	ISO-EF
	CTRL	4 UT (19–26 y), M	NA	–	–	
Tillin et al.[56a]	ISO-RT (KE)	9 UT (21.0 ± 1.0 y), M	ISO KE: 4 sets × 10 rep @ 100 % MVC (3 s per rep) 120 s rest b/w sets	4	4	ISO-KE
Tillin et al.[57a]	EISO-RT (KE)	10 UT (20.0 ± 2.0 y), M	EISO KE: 4 sets × 10 rep @ 95 % MVC (1 s per rep) 120 s rest b/w sets	4	4	ISO-KE
Parker et al.[21]	ISO-RT (KE)	5 UT (27.7 ± 5.7 y), M	ISO KE (UNI): 1 sets × 10–30 rep @ 100 % MVC	3–6	19	ISO-KE
	DYN-RT (KE)	5 UT (27.7 ± 5.7 y), M	DYN KE (UNI): 100–300 rep (load = 63 N)			
Rich et al.[54a]	ISO-RT (KE)	10 MT (22.9 ± 2.1 y), M	ISO KE: 5 sets × 10 rep @ 100 % MVC (4 s per rep)	3	8	ISO-KE
	CTRL	10 T (22.9 ± 2.1 y), M	NA	–	–	
Maeo et al.[19]	ISO-RT (EF)	13 UT (21.8 ± 1.6 y), M	ISO EF & EE: 5 sets × 10 rep @ 100 % MVC (4 s per rep) 120 s rest b/w sets	3	4	ISK-EE & EF
	CTRL	10 UT (21.8 ± 1.6 y), M	NA	–	–	
Tanaka et al.[55]	ISO-RT (PF)	8 UT (21.7 ± 0.7 y), F	3 sets × 20 rep @ 30 % MVC (3 s per rep) 120 s rest b/w sets	3	4	ISO-DF & PF; MT
	CTRL	8 UT (21.88 ± 2.1 y), F	NA	–	–	
Mohr et al.[53]	ISO-RT (KE)	5 UT (21–29 y), F	1 set × 10 rep @ 100 % 1RM (10 s per rep) 10 s rest b/w rep	5	3	ISO-KE
	CTRL	6 UT (21–29 y), F	NA	–	–	
Weir et al.[58a]	ISO-RT (KE)	9 UT (20.0 ± 1.0 y), F	2 sets × 10 rep @ 80 % 1RM (6 s per rep) 120 s rest b/w sets	3	6	ISO-KE

Table 1 (*continued*)

Study	Group	Subjects	Prescription	Frequency	Duration	Assessment/s
Sato et al.[38]	ISO-RT (EF)	13 UT (21.0 ± 0.9 y), M/F	1 sets × 1 rep @ 100 % 1RM (3 s per rep)	5	4	ISO-EF; MT
	ISK-RT(EF)	13 UT (21.4 ± 0.7 y), M/F	One maximal effort from 10° (0.17 rad) to 100° (1.75 rad) at the angular velocity of 30°/s per session			
	CTRL	10 UT(22.2 ± 0.7 y), M/F	NA	–	–	
Oliveira et al.[30]	ISO-RT (KE)	9 UT (23.2 ± 3.8 y), M	ISO KE: 3 sets × 8 rep @ 90 % MVC (5 s per rep) 120 s rest b/w sets	3	6	ISO-KE
	CTRL	9 UT (23.2 ± 3.8 y), M	NA	–	–	
Ohmori et al.[29]	ISO-RT(KE) one day per two weeks	7 UT (19.1 ± 0.6 y), M	ISO KE: 3 sets × 1 rep @ 100 % MVC (5 s per rep), 60 s rest b/w sets	(1/2) × wk-1	8	ISO-KE
	ISO-RT (KE) one day per 3 weeks	7 UT (18.9 ± 0.2 y), M	ISO KE: 3 sets × 1 rep @ 100 % MVC (5 s per rep), 60 s rest b/w sets	(1/3) × wk-1	12	
Maffiuletti et al.[20]	ISO-RT (KE) progressive	8 UT (24.5–4.4 y), M	ISO KE: 6 sets × 6 rep @ 100 % MVC (4 s per rep) 180 s rest b/w sets	3	7	ISO & ISK-KE
	ISO-RT (KE) Ballistic	8 UT(21.6–2.0 y), M	ISO KE: 6 sets × 6 rep @ 100 % MVC (1 s per rep) 180 s rest b/w sets			
	CTRL	5 UT (28.4 ± 3.7 y), M	NA	–	–	
Meyers et al.[52]	ISO-RT (EF) 3bout	10 UT (18–23 y), M	ISO EF: 3 sets × 1 rep @ 100 % MVC (6 s per rep) 30 s rest	3	6	ISO-EF
	ISO-RT (EF) 20bout	10 UT (18–23 y), M	ISO EF: 1 set × 20 rep @ 100 % MVC (6 s per rep) 30 s rest b/w rep			
	CTRL	9 UT (20.1 ± 1.0 y), M	NA	–	–	

BP = bench pull; CON = concentric; CTRL = non-exercise control group; CSA = cross-sectional area; DF = dorsiflexion; DJ = depth jump; DL = deadlift; ECC = eccentric; EE = elbow extension; EF = elbow flexion; EISO = explosive isometric; F = females; HE = hip extension; ISO = isometric; ISK = isokinetic; KE = knee extension; LN = lunge; M = males; MT = muscle thickness; MVC = maximal voluntary contraction; MV = muscle volume; MTP = mid-thigh pull; MT = muscle thickness; NA = not applicable; NR = not reported; PBP = pull bench press; PF = plantar flexion; PLY = plyometric; PU = pull up; REP = repetitions; SQT = squat; SJ = split jump; SLB = single leg bounding; SL-RDL = single leg Romanian Deadlift; SP = shoulder press; SQ = squat; SR = seated row; T = trained; UNI = unilateral; UT = untrained; y = years; sec = seconds.

[a] Indicates studies employed within-subject control group.

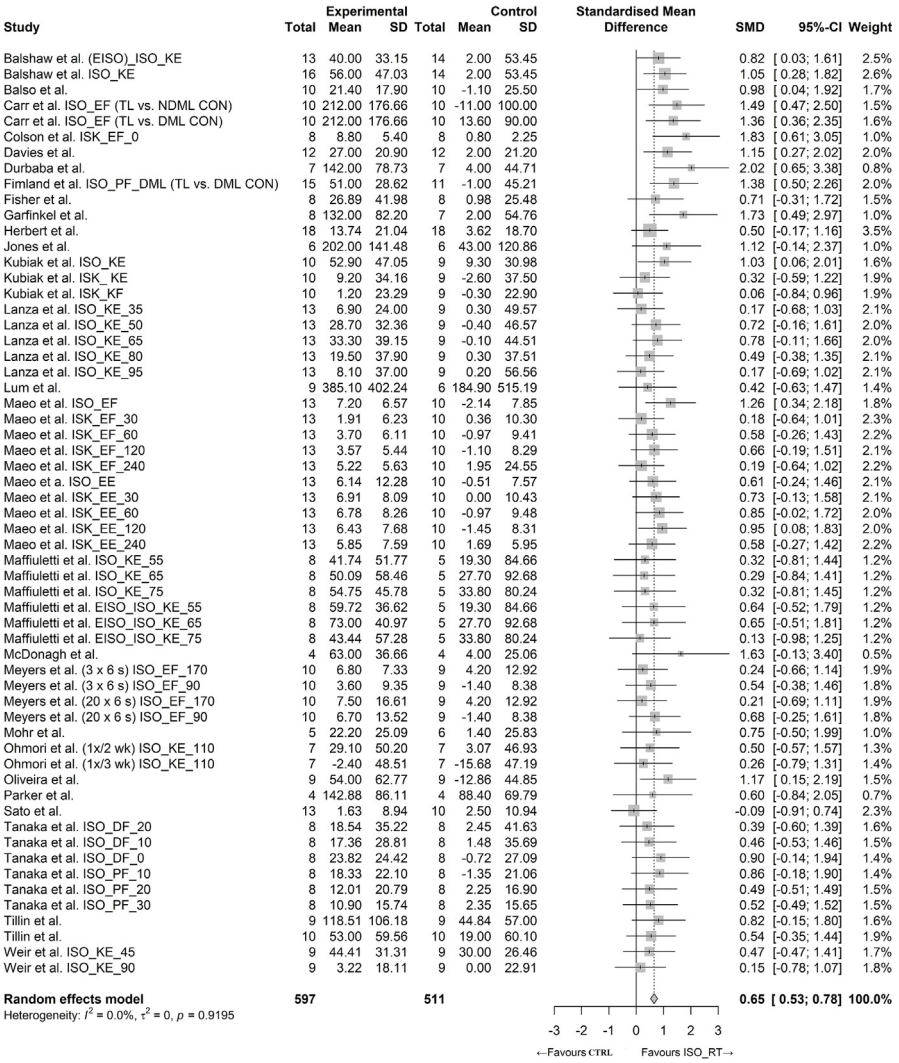

Fig. 2. Meta-analysis forest plot comparing strength gain between ISO-RT and CTRL.

Study	Experimental Total	Experimental Mean	Experimental SD	Control Total	Control Mean	Control SD	SMD	95%-CI	Weight
Follland et al. ISO_KE_50	33	40.66	57.00	33	17.00	48.04	0.44	[-0.05; 0.93]	7.4%
Follland et al. ISO_KE_70	33	66.00	57.00	33	46.00	48.12	0.37	[-0.11; 0.86]	7.5%
Follland et al. ISO_KE_90	33	59.00	71.63	33	50.00	39.04	0.15	[-0.33; 0.64]	7.6%
Follland et al. ISO_KE_110	33	28.67	79.64	33	17.00	40.21	0.18	[-0.30; 0.67]	7.6%
Follland et al. ISK_KE_45	33	26.58	7.05	33	24.02	6.40	0.38	[-0.11; 0.86]	7.5%
Follland et al. ISK_KE_150	33	19.26	4.71	33	16.30	4.70	0.62	[0.13; 1.12]	7.3%
Follland et al. ISK_KE_300	33	11.78	3.20	33	11.63	3.58	0.04	[-0.44; 0.53]	7.6%
Jones et al.	6	202.00	141.48	6	80.00	135.60	0.81	[-0.38; 2.01]	1.2%
Lee et al. (vs. IST)_ISO_KE_30	11	37.20	20.98	10	16.40	22.49	0.92	[0.01; 1.83]	2.1%
Lee et al. (vs. IST)_ISO_KE_50	11	50.10	33.49	10	22.30	32.98	0.80	[-0.09; 1.70]	2.2%
Lee et al. (vs. IST)_ISO_KE_70	11	62.00	44.22	10	34.00	45.16	0.60	[-0.28; 1.48]	2.3%
Lee et al. (vs. IST)_ISO_KE_90	11	77.30	49.72	10	49.00	41.14	0.59	[-0.29; 1.47]	2.3%
Lee et al (vs. IST)_ISK_KE_180	11	15.80	23.56	10	26.60	23.03	-0.44	[-1.31; 0.42]	2.4%
Lee et al. (vs. ISK)_ISO_KE_30	11	37.20	20.98	10	22.50	31.61	0.53	[-0.34; 1.41]	2.3%
Lee et al. (vs. ISK)_ISO_KE_50	11	50.10	33.49	10	20.80	29.02	0.89	[-0.01; 1.80]	2.2%
Lee et al. (vs. ISK)_ISO_KE_70	11	62.00	44.22	10	38.50	29.11	0.60	[-0.28; 1.48]	2.3%
Lee et al. (vs. ISK)_ISO_KE_90	11	77.30	49.72	10	37.90	35.19	0.87	[-0.03; 1.77]	2.2%
Lee et al. (vs. ISK)_ISK_KE_45	11	28.90	38.25	10	48.40	39.97	-0.48	[-1.35; 0.39]	2.3%
Lee et al. (vs. ISK)_ISK_KE_180	11	15.80	23.56	10	28.40	31.26	-0.44	[-1.31; 0.43]	2.4%
Lum et al. (vs. TRT)	9	3.70	4.88	8	0.30	4.43	0.69	[-0.30; 1.68]	1.8%
Lum et al. (vs. PT)	9	3.70	4.88	9	1.30	4.57	0.48	[-0.46; 1.42]	2.0%
Lum et al. ISO_SQ_90	10	226.40	296.14	10	38.10	437.74	0.48	[-0.41; 1.37]	2.2%
Lum et al. ISO_SQ_120	10	248.20	541.08	10	81.30	840.12	0.23	[-0.65; 1.11]	2.3%
Lum et al. ISO_BP_90	10	206.10	269.29	10	59.80	213.68	0.58	[-0.32; 1.47]	2.2%
Lum et al. ISO_BP_120	10	271.20	391.40	10	174.40	257.66	0.28	[-0.60; 1.16]	2.3%
Lum et al. ISO_PBP_90	10	126.30	141.21	10	60.70	203.04	0.36	[-0.53; 1.24]	2.3%
Lum et al. ISO_PBP_120	10	195.60	257.81	10	80.30	248.70	0.44	[-0.45; 1.33]	2.2%
Parker et al.	4	142.88	86.11	5	51.14	139.90	0.68	[-0.70; 2.06]	0.9%
Sato at al.	13	1.63	8.94	13	3.14	7.42	-0.18	[-0.95; 0.59]	3.0%
Random effects model	**453**			**442**			**0.35**	**[0.21; 0.48]**	**100.0%**

Heterogeneity: $I^2 = 0.0\%$, $\tau^2 = 0$, $p = 0.7010$

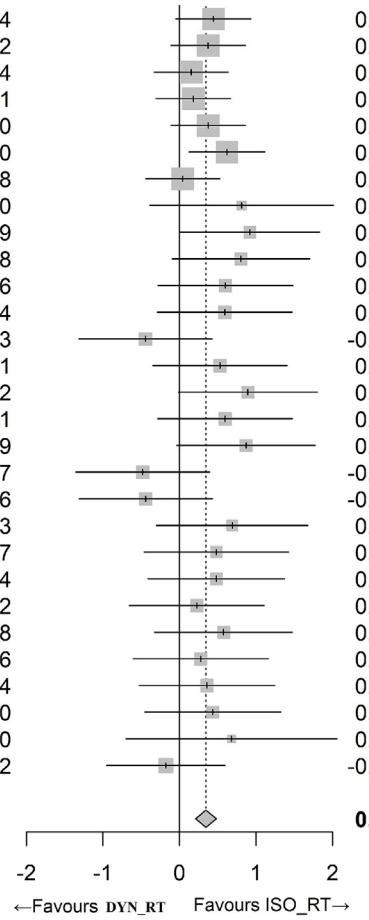

Fig. 3. Meta-analysis forest plot comparing strength gain between ISO-RT and DYN-RT.

(SMD = 0.33, p < 0.0001, 95 % CI = 0.18 to 0.47). For isokinetic strength, the difference between the two groups was not statistically significant (SMD = −0.20, p = 0.24, 95 % CI = −0.55 to 0.14). Although Egger's test detected publication bias (intercept = 1.13, p = 0.002), the Trim and Fill method had minimal impact on the final effect size (SMD = 0.09, p = 0.59, 95 % CI = −0.25 to 0.45).

3.2.1.3. Duration of intervention. For ISO-RT vs. CTRL, both short-term (≤6 weeks) and long-term (>6 weeks) interventions significantly improved strength. Short-term studies yielded an SMD of 0.63 (p < 0.0001, 95 % CI = 0.48 to 0.77), whilst long-term studies had an SMD of 0.67 (p < 0.0001, 95 % CI = 0.42 to 0.93), with no heterogeneity (I^2 = 0 %). Egger's test for short-term studies (intercept = −0.78, p = 0.01) indicated potential publication bias, but the trim-and-fill method adjusted the effect size to 0.49 (p < 0.0001, 95 % CI = 0.35 to 0.62), maintaining significance.

For ISO-RT vs. DYN-RT, long-term studies (SMD = 0.34, p < 0.0001, 95 % CI = 0.19 to 0.49) significantly favoured ISO-RT, with minimal heterogeneity (I^2 = 9 %) and no publication bias (Egger's test: intercept = 0.12, p = 0.35). In short-term studies, the initial effect favoured ISO-RT (SMD = 0.34, p = 0.02, 95 % CI = 0.04 to 0.63), but Egger's test (intercept = −3.32, p = 0.0003) indicated bias. After applying the trim-and-fill method, the effect size decreased to 0.24 (p = 0.06, 95 % CI = −0.01 to 0.50) and lost significance, with no heterogeneity (I^2 = 0 %) before or after adjustment.

3.3. Strength: meta-regression

For ISO-RT vs. CTRL, none of the tested variables significantly moderated muscle strength improvements. Although intensity (MVC%) showed a baseline effect (β = 1.0025, 95 % CI = 0.46 to 1.54, 95 % PI = 0.54 to 0.78), it was not statistically significant (p = 0.19). Similarly, the number of repetitions (β = 0.0072, p = 0.51, 95 % PI = 0.54 to 0.79), time under tension (β = −0.0018, p = 0.90, 95 % PI = 0.53 to 0.78), and rest between sets (β = 0.0005, p = 0.72, 95 % PI = 0.52 to 0.77) had no significant impact on strength gain. For ISO-RT vs. DYN-RT, no variables significantly moderated the effect on muscle strength. Analyses showed nonsignificant effects for intensity (MVC%) (β = 0.09, p = 0.90, 95 % PI = −0.04 to 0.64), repetitions (β = 0.01, p = 0.86 p = 0.90, 95 % PI = 0.11 to 0.46), time under tension (β = 0.01, p = 0.72, 95 % PI = 0.02 to 0.53), and rest between sets (β = 0.00, p = 0.99, p = 0.90, 95 % PI = 0.19 to 0.78).

3.4. Sensitivity analysis

Leave-one-out sensitivity analyses were conducted for both primary comparisons, ISO-RT versus CTRL and ISO-RT versus DYN-RT, to evaluate the stability of the pooled effect estimates for combined isometric and isokinetic strength outcomes. For the ISO-RT vs. CTRL comparison, the SMD remained highly stable across all iterations, ranging from 0.64 to 0.67, with all confidence intervals remaining statistically significant (p < 0.0001). Similarly, for the ISO-RT vs. DYN-RT analysis, SMD

Table 2
Results of the study quality evaluation using the TESTEX.

References	Items												Total points (from a maximum of 15)
	1	2	3	4	5	6	7	8	9	10	11	12	
Balshaw et al.[39]	1	1	1	1	0	0	0	2	1	0	1	1	9
Balso et al.[40]	1	1	1	1	0	1	0	2	1	0	0	1	9
Carr et al.[41]	1	1	1	1	0	0	0	2	1	0	1	1	9
Colson et al.[42]	1	1	1	1	0	0	0	2	1	0	1	1	9
Davies et al.[43]	1	1	1	1	0	3	0	2	1	0	1	1	12
Durababa et al.[44]	1	1	1	1	0	0	0	2	1	0	1	1	9
Oliveira et al.[30]	1	1	1	1	0	0	0	2	1	0	1	1	9
Fimland et al.[45]	1	1	1	1	0	0	0	2	1	0	1	1	9
Fisher et al.[46]	1	0	0	1	0	0	0	2	1	0	0	1	6
Folland et al.[23]	1	0	1	1	0	0	0	2	1	1	1	1	9
Garfinkel and Cafarelli[47]	1	1	1	1	0	0	0	2	1	0	1	1	9
Herbert et al.[48]	1	1	1	1	0	0	0	2	1	0	1	1	9
Jones et al.[24]	1	1	1	1	0	0	0	2	1	1	1	1	10
Kubiak et al.[49]	1	1	1	1	0	0	0	2	1	0	1	1	9
Lanza et al.[18]	1	1	1	1	0	0	0	2	1	0	1	1	9
Lee et al.[22]	1	1	1	1	0	0	0	2	1	1	1	1	10
Lum et al.[50,59]	1	1	1	1	0	0	0	2	1	1	1	1	10
Lum et al.[15]	1	1	1	1	0	0	0	2	1	1	1	1	10
Lum et al.[17,60]	1	1	1	1	0	0	0	2	1	0	1	1	9
Maeo et al.[19]	1	1	1	1	0	0	0	2	1	0	1	1	9
Maffiuletti and Martin[20]	1	1	1	1	0	0	0	2	1	0	1	1	9
McDonagh et al.[51]	1	1	1	1	0	0	0	2	1	0	1	1	9
Meyers et al.[52]	1	1	1	1	0	0	0	2	1	0	1	1	9
Mohr et al.[53]	1	1	1	0	0	0	0	2	1	0	1	1	8
Ohmori et al.[29]	1	1	1	1	0	0	0	2	1	0	1	1	9
Parker[21]	1	1	1	1	0	0	0	2	1	1	1	1	10
Rich et al.[54]	1	1	1	0	0	0	0	2	1	0	1	1	8
Sato et al.[38]	1	1	1	1	0	0	0	2	1	1	1	1	10
Tanaka et al.[55]	1	1	1	1	0	0	0	2	1	0	1	1	9
Tillin et al.[56]	1	1	1	1	0	0	0	2	1	0	1	1	9
Tillin et al.[57]	1	1	1	1	0	0	0	2	1	0	1	1	9
Weir et al.[58]	1	1	1	1	0	0	0	2	1	0	1	1	9

Note for items 1 through 12: (1) eligibility criteria specified (1 point); (2) randomisation defined (1 point); (3) allocation concealment (1 point); (4) groups similar at baseline (1 point); (5) assessor blinding in study reporting (1 point); (6) outcome measures assessed in 85 % of patients (3 points); (7) intention-to-treat analysis (1 point); (8) between-group statistical comparisons reported (2 points); (9) point measures and measures of variability for all reported outcome measures (1 point); (10) activity monitoring in the controlled group (1 point); (11) relative exercise intensity remained constant (1 point); and (12) exercise volume and energy expenditure (1 point).

values ranged narrowly from 0.33 to 0.37, again with all iterations yielding significant results ($p < 0.0001$). In both cases, no single study had undue influence on the overall findings, reinforcing the robustness and reliability of the meta-analytic estimates. These results support the conclusion that ISO-RT consistently improves muscular strength relative to non-training controls and DYN-RT across diverse study designs and populations.

3.5. Study quality analysis

Using the TESTEX scale, the mean total score for study quality was 9.1 (median 9.0) out of a possible 15 points, with scores ranging from 6 to 12. Only one study achieved an excellent quality score of 12,[43] whilst most studies were classified as good quality, scoring between 8 and 11. All studies satisfied essential criteria, including specifying eligibility criteria, providing point measures and variability for reported outcomes, maintaining consistent relative exercise intensity, and reporting exercise volume and energy expenditure. However, none of the studies implemented intention-to-treat analysis or blinding of assessors. In studies with DYN-RT, activity monitoring was included, whereas studies with CTRL did not monitor participant activity. The quality analysis of each study is illustrated in Table 2.

4. Discussion

To our knowledge, this meta-analysis is the first to directly compare the effects of ISO-RT on muscular strength relative to both DYN-RT and a CTRL group. The findings indicate that ISO-RT significantly improves strength compared to CTRL (moderate effect) and shows modest but favourable strength improvements compared to DYN-RT (small effect). Subgroup analyses revealed that ISO-RT elicited greater improvements in isometric strength than DYN-RT (small effect), whilst isokinetic strength did not significantly differ between the groups. These findings challenge the common perception that ISO-RT is limited to joint-angle specific improvements, suggesting broader applicability across different strength assessments.[61]

4.1. Isometric and isokinetic strength adaptations

This meta-analysis revealed that ISO-RT significantly improved strength compared to CTRL, and demonstrated modest but favourable effects over DYN-RT. Notably, ISO-RT elicited greater gains in isometric strength, whilst isokinetic strength improvements were similar between ISO-RT and DYN-RT. These findings support the principle of Specific Adaptations to Imposed Demands (SAID), where strength gains tend to be specific to the contraction mode trained. The ability of ISO-RT to generate higher peak forces than concentric actions may explain its effectiveness, especially when training is conducted at longer muscle lengths. Longer muscle length ISO-RT is associated with greater neuromuscular activation and mechanical tension, which have been shown to enhance both absolute strength and transfer to dynamic performance.[8,62]

Strength improvements following ISO-RT were consistently observed across both upper and lower body assessments when compared to CTRL, regardless of intervention duration. When compared to DYN-RT, ISO-RT produced more favourable results, especially for lower body and isometric strength. In contrast, differences in isokinetic strength between ISO-RT and DYN-RT were small and not statistically significant. Longer intervention durations tended to yield greater improvements, whilst short-term interventions showed more variable results, especially when accounting for potential bias. Overall, these findings highlight the effectiveness of ISO-RT, particularly when implemented over longer durations, as a viable strategy for enhancing muscular strength.[63–65] Seven studies[21–24,38,59,60] compared ISO-RT and DYN-RT using strength measures across varying joint angles and velocities. Whilst only three comparisons favoured DYN-RT, the greatest strength transfer in ISO-RT was observed when training at longer muscle lengths (e.g. 90–110° of knee flexion[23]), where passive tension and muscle stretch are maximised. These results suggest that joint-angle selection may play an important role in enhancing strength transfer from ISO-RT to dynamic performance.

4.2. Effect of multi-angle isometric training

Although most ISO-RT studies used a single joint angle, multi-angle protocols may offer additional benefits. One proposed mechanism is the ability to strengthen “sticking points” within the range of motion.[66] These are positions of mechanical disadvantage commonly encountered in compound lifts such as the bench press, squat, and deadlift.[66,67] Strengthening these positions may enhance overall force output, improve movement efficiency, and support better transfer to dynamic performance. However, caution is warranted, as only one included study applied ISO-RT across multiple joint angles, and its within-group design limits generalizability. Despite the limited empirical evidence, the use of multi-angle ISO-RT is conceptually supported by practitioners and warrants further investigation to determine the most effective joint angle combinations and contraction parameters.

Relatedly, findings from Spitz et al.[65] support the concept of cross-transfer between muscle actions, showing that ISO-RT can improve dynamic performance under certain conditions. Whilst the smaller effects observed in isokinetic compared to isometric tests reinforces the SAID principle, they also indicate that ISO-RT can confer partial improvements in non-isometric tasks, particularly when performed under

conditions (e.g. long muscle lengths or multi-angle formats) that favour transfer.

4.3. Effect of training variables

Although ISO-RT allows for manipulation of several variables, such as intensity, time under tension, joint angle, and volume, our meta-regression did not identify any of these as significant moderators of strength outcomes when comparing ISO-RT to DYN-RT and CTRL. This contrasts with previous reviews, including Oranchuk et al.,[8] which proposed training volume as a key factor influencing strength adaptations to isometric training. This lack of significance may reflect methodological limitations in the included studies, such as small sample sizes, inconsistent reporting of key parameters, and limited variation in training prescriptions. It may also suggest that strength outcomes in ISO-RT are influenced by complex interactions between variables, rather than any single factor in isolation. As such, future studies should aim for greater standardisation and more detailed reporting to clarify the role of these variables in modulating ISO-RT effectiveness.

4.4. Strengths and limitations

A key strength of this meta-analysis was the inclusion of time-matched DYN-RT or CTRL groups, which improves the internal validity of comparisons and allows for clearer attribution of effects to the training interventions allowing for a more accurate comparison of muscle strength outcomes. Unlike an earlier review that assessed pre-to-post changes without controls,[8] this design helps mitigate confounding factors such as test learning effects and natural progression over time.[64] Another strength of this meta-analysis was the inclusion of both isometric and isokinetic strength outcome measures, which provided a comprehensive picture of strength adaptations and allowed exploration of cross-transfer effects between contraction modes.

There are several limitations that must be acknowledged. Most studies involved untrained participants, which limits the generalizability of findings to trained individuals or athletic populations. Furthermore, inconsistent and incomplete reporting of critical variables, such as joint angles, muscle lengths, intensity, frequency, and rest intervals, restricted the ability to conduct detailed subgroup or moderator analyses. In addition, only one study applied a multi-angle ISO-RT protocol, limiting the strength of conclusions regarding its effectiveness. Future research should aim to standardise training prescriptions, include trained cohorts, and explicitly report joint angles and muscle lengths, especially given emerging evidence supporting the benefits of training at longer muscle lengths. Finally, although our analyses showed low statistical heterogeneity ($I^2 = 0\,\%$), this likely reflects homogeneity in study design and population characteristics, rather than true methodological consistency. Thus, there remains a need for broader research across diverse contexts and populations to confirm and extend these findings.

4.5. Conclusion

This systematic review and meta-analysis found that ISO-RT significantly improves muscular strength compared to no training and demonstrates comparable or superior performance to DYN-RT in certain contexts. ISO-RT was particularly effective in enhancing isometric strength, with some evidence of transfer to isokinetic and potentially dynamic performance when performed at longer muscle lengths. These results highlight ISO-RT as a viable and potentially underutilised strategy for improving strength across different joint angles and muscle actions.

CRediT authorship contribution statement

MG and DH conceived the idea for the review. MG, JL, and MK conducted a search, selection of studies, data extraction, and quality assessment. FE conducted the analysis. MG and DH interpreted the data. MG and DH drafted the initial manuscript. AS and CF contributed to writing the manuscript. All authors revised critically, reviewed and approved the final manuscript. The authors have no conflict of interest to declare.

Confirmation of ethical compliance

No ethics is required for this review.

Funding information

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Declaration of interest statement

The authors declare no conflict of interest.

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None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jsams.2025.07.011.

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